

Week 4 – Data Visualization with Matplotlib and Seaborn

Objective

The objective of Week 4 was to learn how to visualize data using Python libraries such as Matplotlib and Seaborn. Data visualization helps in understanding patterns, relationships, and trends within datasets. It is an important step in data analysis and helps present data in a meaningful and visual format.

1. Matplotlib

Matplotlib is a widely used Python library for creating basic data visualizations. It allows users to generate different types of plots such as line graphs, bar charts, histograms, and scatter plots.

Example:

```
import matplotlib.pyplot as plt
```

```
plt.plot([1,2,3], [4,5,6])  
plt.title("Simple Plot")  
plt.show()
```

2. Seaborn

Seaborn is a data visualization library built on top of Matplotlib. It provides more advanced and visually appealing statistical plots.

Example:

```
import seaborn as sns
```

```
sns.heatmap([[1,2],[3,4]])
```

3. Data Visualization Techniques

Histogram

A histogram is used to show the distribution of numerical data.

Scatter Plot

A scatter plot shows the relationship between two variables.

Heatmap

A heatmap displays correlations between variables using colors.

Pairplot

A pairplot shows pairwise relationships between multiple variables.

Program Implemented – Data Visualization Dashboard

Description

A Python script was developed to visualize a dataset using different plots. The dataset contains features such as age, study hours, and scores. Various plots were created to analyze relationships and patterns in the data.

Program Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Sample dataset
data = {
    "Age": [18, 19, 20, 21, 22, 23, 24],
    "Study_Hours": [2, 3, 4, 5, 6, 4, 3],
    "Score": [60, 65, 70, 75, 80, 78, 72]
}

df = pd.DataFrame(data)

# Histogram
plt.hist(df["Score"])
plt.title("Score Distribution")
plt.xlabel("Score")
plt.ylabel("Frequency")
plt.show()

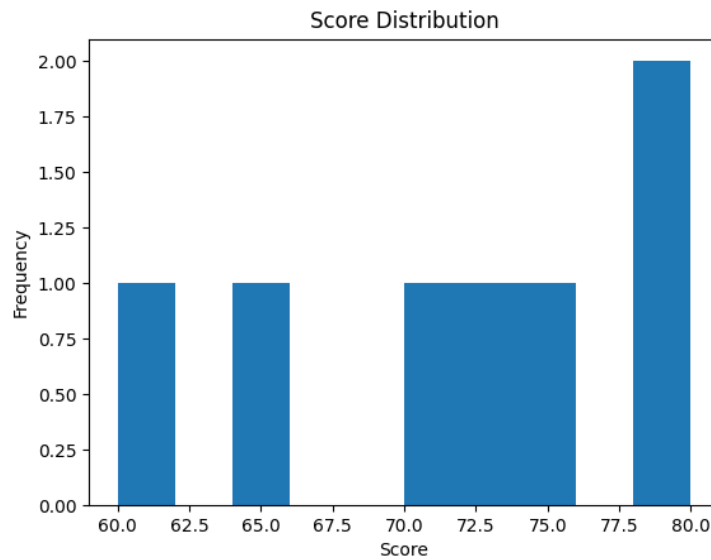
# Scatter Plot
plt.scatter(df["Study_Hours"], df["Score"])
plt.title("Study Hours vs Score")
plt.xlabel("Study Hours")
plt.ylabel("Score")
plt.show()
```

```
# Heatmap
sns.heatmap(df.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

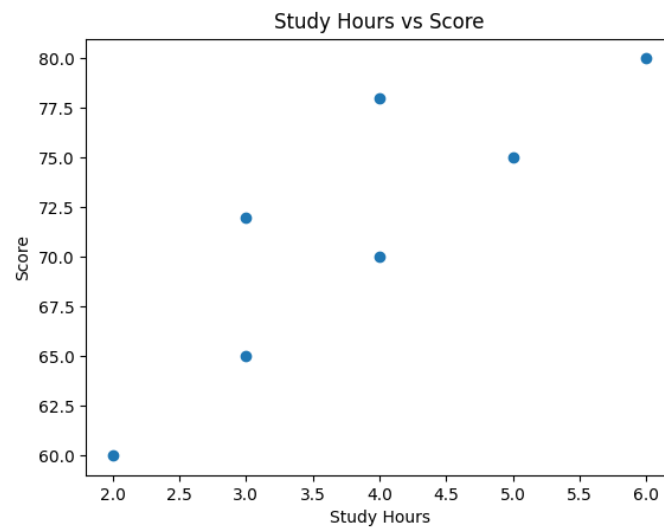
```
# Pairplot
sns.pairplot(df)
plt.show()
```

OUTPUT:

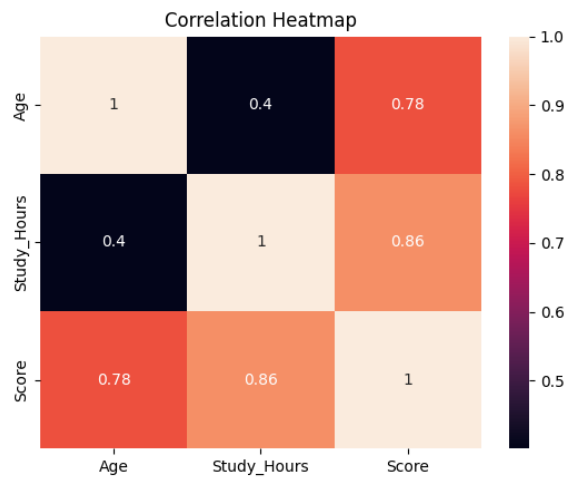
Histogram



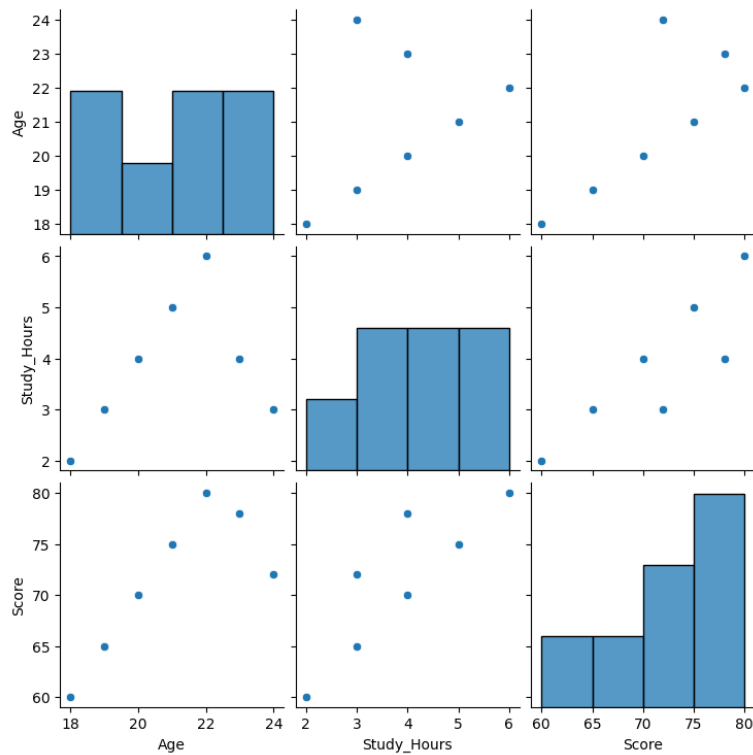
Scatter plot



Heatmap



Pair plot



Conclusion

Week 4 helped in understanding how to visualize data using graphs and charts. It provided practical knowledge of using Matplotlib and Seaborn to analyze relationships and patterns in datasets.